

SIMATS ENGINEERING

Department of Computer Science and Engineering



SIMATS
Sri Sankaranarayanan Institute of Medical and Technical Sciences
(Deemed to be University, under Section 3 of UGC Act 1956)

CSA15 Cloud Computing and Big Data Analytics

CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense

Editable student template for application-based cloud virtualization and SDDC/cloud operations defense

Assessment Metadata

Course Code	CSA1512
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense
Submission Type	Individual digital case analysis and infrastructure defense based on the same capstone title used in CO1, CO2 AT1 and CO2 AT2
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed DOCX template and GitHub repository link through Google Classroom

How This Template Works

- Use the same capstone title used in CO1, CO2 AT1 and CO2 AT2.
- Do not recalculate all VM sizing or repeat all configuration screenshots. Summarize the final AT1/AT2 evidence and use it for design defense.
- Answer all five questions with application-oriented reasoning connected to the capstone product.
- Include sustainable cloud operation decisions wherever relevant: right-sized resources, log retention, storage lifecycle, scheduled shutdown, API-call reduction and VM/container/serverless comparison.
- Paste the final topology diagram in the given response area. The sample diagram is only a structure; redraw it for the student's own product.
- Upload this completed document and supporting evidence to GitHub, then submit the repository link through Google Classroom.

Assessment Focus

AT1 answered what VM is required. AT2 showed configuration or simulation evidence. AT3 asks whether the student can defend the complete infrastructure design under application, security, operations and failure conditions.

Student and Capstone Details

Student Name	T.PRAKEERTHI
Register Number	192524279

Class / Slot	Slot B
Capstone Title	Medi Queue AI: Hospital of Queue and No-show. prediction App.
CO1 Evidence Link	
CO2 AT1 VM Recommendation	vCPU: ____ RAM: ____ GB Storage: ____ GB VM Class: ____
CO2 AT2 Evidence Link	
GitHub Repository Link	tarigondaprakeerthi/CSA1512-Cloud-computing
Submission Date	30/07/26

Evidence Carry-Forward Summary

Evidence Source	Student Entry	How It Supports AT3 Defense
CO1 concept map / presentation		
CO2 AT1 sizing workbook		
VM calibration sheet		
CO2 AT2 practical evidence		
Capstone architecture decision		

Q1: Virtualization Value and Capstone Cloud Deployment Summary

Explain your capstone product, expected workload, cloud need and deployment context. Analyze how virtualization improves cost efficiency, resource utilization, workload isolation, infrastructure flexibility and cloud deployment readiness for your product.

Design Point	Student Response
Capstone product and users	MediQueue AI is a cloud-based hospital appointment and OP queue management system. Users include patients, doctors, hospital staff, and administrators .
Expected workload	Around 500–2,000 users per day with increased traffic during morning OP hours. The system should support multiple hospitals and many simultaneous appointment requests.
Cloud need	Cloud provides 24/7 access, secure storage, real-time synchronization, AI prediction, notifications, and scalability for handling many users.
Virtualization value	Virtual machines allow the application to run in isolated environments , making deployment, testing, maintenance, and scaling easier without affecting other services.
Cost and utilization benefit	Cloud follows a pay-as-you-use model, reducing hardware costs. Resources can automatically scale up during peak

	hours and scale down during low usage, improving utilization and lowering expenses.
Isolation and flexibility benefit	Each service (web application, AI model, database) can run independently, improving security, reliability, and fault isolation . Resources can be upgraded or modified easily without interrupting the entire system.
Case Analysis Response: Write 8-12 technical lines connecting virtualization value to your capstone product.	Case Analysis Response Technical Lines: Virtualization is a key technology used in the MediQueue AI cloud platform. It allows multiple virtual machines to run on a single physical server, improving resource utilization. The web application, AI prediction engine, and database can be deployed on separate virtual machines for better performance. Each virtual machine operates independently, preventing failures in one service from affecting others. Virtualization enables quick allocation of CPU, memory, and storage during peak hospital appointment hours. It supports easy scaling as the number of patients and hospitals increases. Virtual machines can be backed up, restored, or migrated with minimal downtime, ensuring service availability. The isolated environment improves security by protecting sensitive patient and appointment data. It also reduces infrastructure costs by maximizing hardware utilization through resource sharing. Thus, virtualization makes MediQueue AI reliable, scalable, secure, and cost-effective for cloud-based healthcare services.

Q2: Rapid Provisioning, Testing and Virtual Machine Lifecycle Case

Using your CO2 AT1 VM recommendation and CO2 AT2 practical evidence, explain how virtual machines, VM images, templates, snapshots, clones or containers support rapid provisioning, testing, deployment, rollback and maintenance for your capstone product.

Lifecycle Element	AT1/AT2 Evidence Used	Capstone Use / Justification
VM image or base environment	Ubuntu 22.04 LTS VM created in Azure (AT1)	Provides a stable Linux environment to host the MediQueue AI web application, AI services, and database.
Template or repeatable setup	Azure VM configuration (Resource Group, VM Size, Network settings) documented in AT1	The same configuration can be reused to deploy additional hospital servers quickly and consistently.
Snapshot / rollback	VM snapshot before software installation and updates	If an update causes issues, the VM can be restored to the previous working state, reducing downtime.

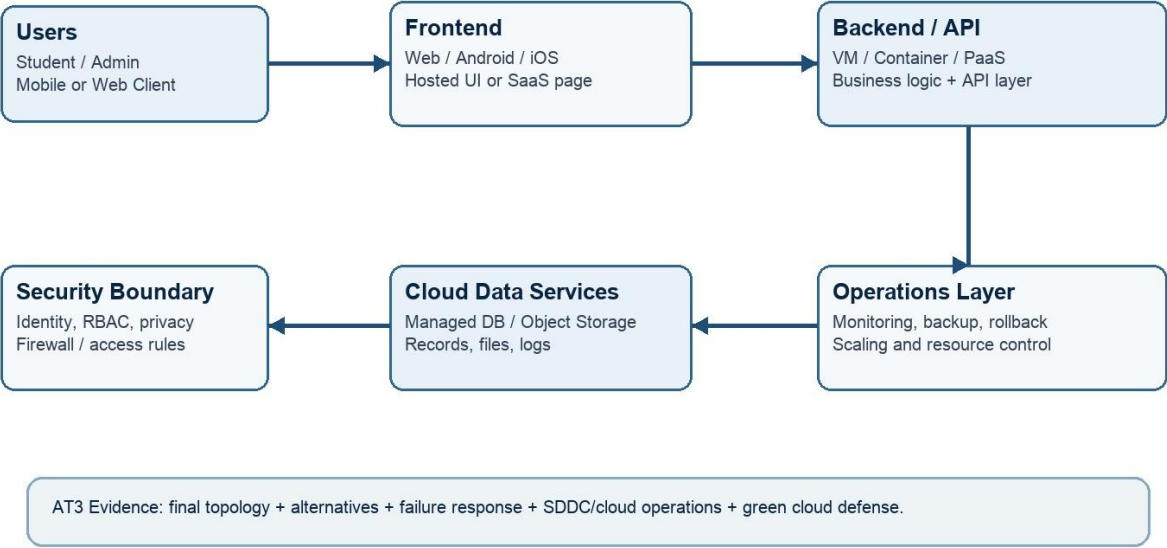
Clone / testing copy	Cloned VM created for testing (AT2)	A duplicate environment is used to test new AI models and application updates without affecting the live system.
Container or VM decision	Virtual Machine selected instead of containers	A VM provides a complete operating system, making it easier to manage the full healthcare application and database securely for the capstone project.
Maintenance and update plan	Regular OS updates, security patches, backups, and monitoring performed in Azure	Ensures MediQueue AI remains secure, reliable, and available while protecting patient data and improving system performance.
Case Analysis Response: Explain how lifecycle practices reduce deployment delay and risk.	Case Analysis Response Lifecycle practices reduce deployment delay by using standardized VM images and templates, allowing the MediQueue AI application to be deployed quickly without repeating manual configuration. Snapshots and rollback features enable fast recovery if an update or deployment fails, minimizing downtime. Cloned virtual machines provide safe testing environments where new features and AI models can be validated before production deployment. Regular maintenance, security patches, and automated backups improve system stability and protect patient data. These practices reduce configuration errors, lower operational risk, and ensure reliable service. As a result, MediQueue AI can be updated faster, recover quickly from failures, and provide continuous healthcare services with minimal interruption.	

Q3: Final Cloud Deployment Topology and SOA/API Service Integration Case

Draw and explain the final deployment topology for your capstone product. Show frontend, backend/API, database, storage, authentication, notification, analytics and any VM/container/managed cloud service components. Explain how SOA/API-based integration connects these modules and supports modular cloud service design.

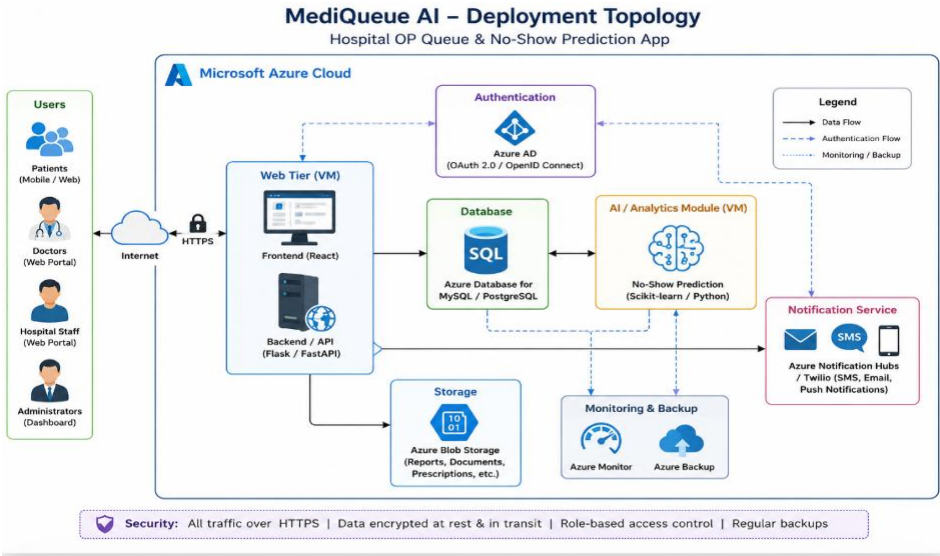
Capstone Cloud Deployment Topology - Sample Structure

Students should redraw this with their own modules, services, VM/container choices and security boundaries.



Architecture Component	Technology / Service	VM / Container / Managed Service / SaaS	Integration Method
Frontend	React.js / HTML, CSS, JavaScript	VM	Accesses backend through HTTPS REST APIs
Backend/API	Python (Flask/FastAPI)	VM	REST API connected to frontend and database
Database	MySQL / PostgreSQL	Managed Service	SQL connection through backend API
Storage	Azure Blob Storage	Managed Service	Secure cloud storage using Azure SDK/API
Authentication	Azure Active Directory (Azure AD)	SaaS	OAuth 2.0 / OpenID Connect authentication
Notification	Azure Notification Hubs / Twilio SMS	Managed Service	API integration for SMS and push notifications
Analytics / AI module	Python, Scikit-learn / TensorFlow	VM	Backend sends appointment data to AI model through API and receives no-show prediction

Paste or draw the final capstone deployment topology diagram here.

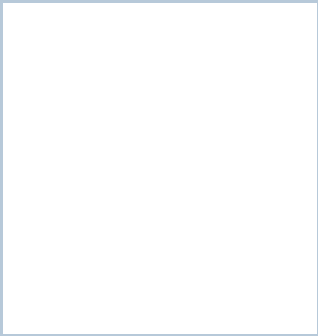


Q4: Design Alternatives, Identity, Access and Federated Cloud Privacy Case

Compare at least two deployment alternatives for your capstone product, such as single VM, multi-tier VM, containerized deployment, managed backend, federated cloud or hybrid cloud model. Justify the selected design by analyzing identity management, role-based access control, federated access, data sharing, privacy protection and security risks.

Deployment Alternative	Advantages	Limitations / Risks	Identity and Privacy Impact
Alternative 1	Full control over hardware and data, works without relying on cloud provider.	High installation and maintenance cost, limited scalability, requires dedicated IT staff.	Patient data stays within the hospital, but security depends entirely on the hospital's infrastructure and management.
Alternative 2	Scalable, cost-effective, high availability, automatic backup, disaster recovery, and easy remote access.	Requires a stable internet connection and depends on cloud service availability.	Supports secure authentication, encrypted data storage, role-based access control, and compliance with healthcare privacy requirements.
Selected design	Combines scalability, reliability, AI integration, secure storage, and easy maintenance. Supports multiple hospitals and users efficiently.	Requires proper cloud security configuration and continuous monitoring	Uses Azure Active Directory for identity management, HTTPS encryption, role-based access control, and encrypted storage to

		to prevent unauthorized access.	protect sensitive patient information.
Security Area		Student Decision	
User roles	Patient, Doctor, Hospital Staff, and Administrator. Each user has different permissions based on their role.		
Authentication method	Secure login using Azure Active Directory (Azure AD) with OAuth 2.0/OpenID Connect and strong passwords (optionally Multi-Factor Authentication).		
Role-based access control	Patients can manage their appointments, doctors can view patient schedules, staff can manage queues, and administrators can manage the entire system.		
Federated access / partner access	Azure AD allows secure login for partner hospitals and authorized healthcare organizations using federated identity.		
Sensitive data handled	Patient personal details, appointment records, medical information, doctor schedules, and AI no-show prediction data.		
Privacy protection decision	Encrypt data in transit (HTTPS) and at rest, implement role-based access control, maintain audit logs, perform regular backups, and comply with healthcare data privacy requirements to protect patient information.		
Case Analysis Response: Defend why the selected design is better for identity, access and privacy.	Case Analysis Response The selected design uses Azure Active Directory , role-based access control (RBAC) , and HTTPS encryption to provide strong identity and privacy protection for MediQueue AI . Each user—patient, doctor, hospital staff, and administrator—is authenticated securely and is granted only the permissions required for their role. This prevents unauthorized access to sensitive patient records. All data is encrypted during transmission and while stored in the cloud, reducing the risk of data theft. Audit logs and regular backups improve accountability and support recovery if a security incident occurs. Compared with a traditional on-premises system, the cloud-based design offers centralized identity management, easier access control, and stronger privacy protection while maintaining a secure and reliable healthcare service.		



Q5: Software-Defined Datacenter and Cloud Operations Defense Case

Analyze how your infrastructure will respond to operational and failure scenarios such as VM failure, database growth, traffic spike, credential leakage, service outage, resource contention or cost increase. Explain how SDDC concepts and cloud operations practices such as automated provisioning, software-defined networking, software-defined storage, monitoring, logging, backup, rollback, scaling, resource control and maintenance improve the final infrastructure design. Include Green Computing decisions such as resource right-sizing, storage lifecycle control, log retention, API-call reduction, scheduled shutdown, VM/container/serverless comparison and carbon/energy proxy indicators where relevant.

Operational / Failure Scenario	Expected Impact	SDDC / Cloud Operations Response	Evidence or Control
VM failure	Application becomes unavailable temporarily.	Restart the VM or recover from a backup/snapshot; use a standby VM if available.	Azure VM monitoring, snapshots, and backup recovery.
Database growth	Slower queries and reduced application performance.	Increase database storage and optimize indexes using managed database scaling.	Database monitoring, automated backups, and storage alerts.
Traffic spike	High response time and possible service slowdown.	Auto-scale compute resources or deploy additional VM instances.	Azure Monitor, load balancing, and autoscaling policies.
Credential leakage	Unauthorized access to patient data and services.	Reset credentials immediately,	Azure Active Directory, RBAC, audit logs, and MFA.

		enable Multi-Factor Authentication (MFA), and revoke compromised sessions.	
Service outage	Appointment booking and notifications become unavailable.	Restore services from backup or switch to a healthy instance in another availability zone.	Health monitoring, backups, and disaster recovery plan.
Resource contention	Reduced system performance due to CPU or memory shortages.	Increase VM resources or distribute workloads across multiple VMs.	Performance monitoring dashboards and resource alerts.
Cost increase	Higher monthly cloud expenses.	Monitor resource usage, stop unused VMs, and resize resources when demand is low.	Azure Cost Management, budgets, and usage reports.

Operational Readiness Area	Student Plan
Automated provisioning	Use Azure Resource Manager (ARM) templates or Infrastructure as Code (IaC) to automatically deploy virtual machines, networking, and storage.
Software-defined networking	Configure Azure Virtual Network (VNet), Network Security Groups (NSGs), and Load Balancer to provide secure and reliable communication between services.
Software-defined storage	Use Azure Blob Storage and Azure Managed Database with automatic storage expansion and encryption.

Monitoring and logging	Use Azure Monitor and Log Analytics to track VM health, application performance, errors, and security events with alerts.
Backup and rollback	Schedule automatic VM and database backups, create snapshots before updates, and restore quickly if failures occur.
Scaling and elasticity	Enable auto-scaling of compute resources during peak hospital hours and scale down during low-demand periods to reduce costs.
Resource control and maintenance	Monitor CPU, memory, and storage usage, apply regular security patches, update software, and stop unused resources to optimize performance.
Green Computing and sustainability control	Reduce energy consumption by shutting down idle VMs, selecting appropriate VM sizes, using auto-scaling, and optimizing resource utilization to lower power usage and cloud costs.
Green Cloud Decision	Student Justification
VM right-sizing from CO2 AT1	Select a small VM (2 vCPU, 4 GB RAM) that matches the application's workload instead of an oversized server. This reduces energy consumption and cloud costs while providing sufficient performance.
VM vs container vs serverless choice	Use containers because MediQueue AI consists of lightweight API services that start quickly, use fewer resources than full VMs, and are easier to scale, reducing overall energy usage.
Scheduled shutdown / idle-resource control	Automatically shut down development and testing environments outside college/project hours (for example, at night and weekends) to avoid wasting compute resources.
Log and file retention policy	Keep application logs for 30 days for troubleshooting, then automate unnecessary storage usage.
Storage lifecycle and backup cleanup	Move old patient reports and backups to lower-cost archive storage after 90 days , and delete outdated backups after the retention period to reduce storage consumption.
API-call or polling reduction	Use push notifications or event-based updates instead of frequent polling by the mobile app. This reduces unnecessary network traffic, server processing, and energy consumption.
Carbon/energy proxy indicator to track	Monitor CPU utilization, memory usage, VM uptime, storage usage, and number of API requests . Lower idle time and optimized resource utilization indicate reduced energy consumption and a smaller carbon footprint.

Final Infrastructure Defense Statement: State whether the design is ready for prototype, pilot or production. Defend reliability, security, SDDC/cloud operations and Green Computing improvements required next.	Final Infrastructure Defense Statement Deployment Status: Prototype Ready The MediQueue AI infrastructure is ready for the prototype stage. The solution provides a secure and scalable cloud environment for hospital appointment booking, queue management, and AI-based no-show prediction. Containers and right-sized cloud resources ensure efficient deployment while keeping infrastructure costs low. The design demonstrates reliability through automated backups, health monitoring, logging, and the ability to scale services as user demand increases. Security is addressed using HTTPS encryption, role-based access control, secure authentication, and regular backup policies to protect patient information. SDDC/Cloud operations are supported with automated provisioning, software-defined networking, software-defined storage, monitoring, logging, and scheduled maintenance for efficient resource management. To further improve Green Computing , the next steps include enabling automatic shutdown of idle resources, optimizing VM/container sizing, reducing unnecessary API calls, implementing storage lifecycle policies, and monitoring energy-related indicators such as CPU utilization and VM uptime. These improvements will reduce power consumption, operational costs, and the application's environmental impact while preparing the infrastructure for future pilot and production deployment.
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CO2 Reflection and Individual Defense

This section must be individually written. It should show what the student understands from the complete CO2 chain: infrastructure sizing, practical configuration evidence and final operations defense.

Reflection Prompt	Student Response
What is the strongest infrastructure decision in your capstone design?	Using containers with cloud hosting is the strongest decision because it provides scalability, efficient resource utilization, and easy deployment while reducing infrastructure costs.
Which CO2 concept became clearer after AT1, AT2 and AT3?	The importance of VM right-sizing and Green Computing became much clearer. Choosing only the required resources reduces cost, energy consumption, and environmental impact.
Which risk is most serious for your capstone: failure, cost, privacy, scaling or sustainability?	Privacy is the most serious risk because MediQueue AI handles sensitive patient information that must be protected using encryption, secure authentication, and access control.
What evidence would you show during viva to defend your infrastructure design?	I would show the cloud architecture diagram, VM/container configuration, monitoring dashboard, backup policy, security settings (HTTPS and authentication), application screenshots, and testing results to demonstrate reliability and cloud readiness.
What will you improve before final capstone deployment?	I will implement auto-scaling, stronger security (MFA and regular security audits), disaster recovery testing, automated monitoring and alerts, and additional Green

	Computing measures such as scheduled shutdown and storage lifecycle management before production deployment.
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Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	The answer uses the same capstone title and connects CO1, CO2 AT1 and CO2 AT2 evidence.
Virtualization and lifecycle reasoning	VM, image, snapshot, clone, container or managed-service decisions are technically justified.
Topology and integration clarity	Architecture diagram and SOA/API explanation show clear module interaction.
Identity, access and privacy	Roles, authentication, RBAC, federated access, sensitive data and privacy decisions are defended.
SDDC/cloud operations	Failure response, monitoring, logging, scaling, backup, rollback and resource control are addressed.
Green Computing defense	Right-sizing, retention, API efficiency, idle-resource control and sustainable cloud choices are explained.
Submission quality	GitHub, template completion, evidence references and individual reflection are complete.

GitHub and Google Classroom Submission

Repository Item	Expected Content
README.md	Capstone title, student details, final VM recommendation, deployment topology summary and AT3 evidence index.
Completed AT3 document	This completed editable DOCX or exported PDF if instructed.
Diagram image	Final topology diagram if created separately.
Evidence references	Links or screenshots from CO1, CO2 AT1 and CO2 AT2 as needed.
Green Computing note	Short README section explaining sustainable cloud choices and resource-control decisions.
Google Classroom	Submit the GitHub repository link through GCR.

Student Assurance

I confirm that this case analysis, design reasoning, topology diagram and final defense statement were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	T.PRAKEERTHI
Register Number	192524279
Signature	T.PRAKEERTHI
Date	31/07/2026

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	